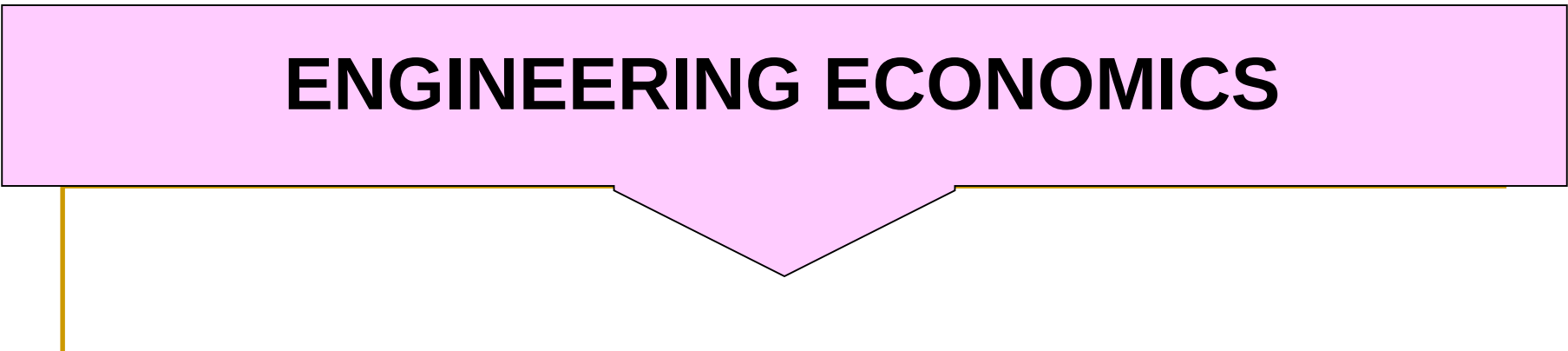
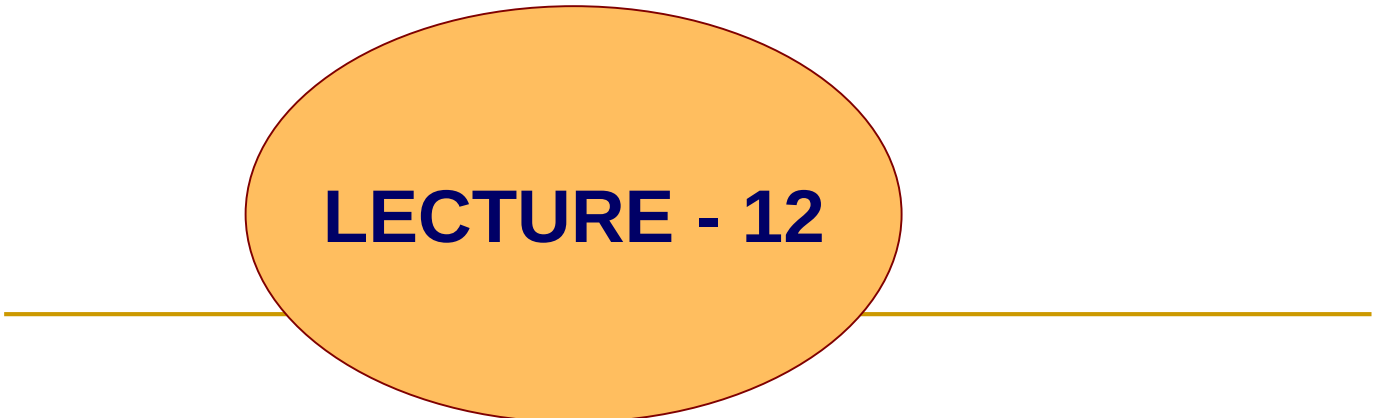


ENGINEERING ECONOMICS



LECTURE - 12



Depreciation

Depreciation may be defined as the decrease in the value of physical assets with the passage of time as a result of wear, deterioration and technological obsolescence.

Why Do We Consider Depreciation?

It is used in the books of accounts for preparing a balance sheet of assets.

Depreciation is viewed as a part of business expenses that reduce taxable income.

Depreciation Example

You purchased a car worth \$15,000 at the beginning of year 2004.

End of Year	Market Value	Loss of Value
0	\$15,000	
1	10,000	\$5,000
2	8,000	2,000
3	6,000	2,000
4	5,000	1,000
5	4,000	1,000

Depreciation

Factors to Consider in Asset Depreciation

- ❑ Depreciable life (how long?)
- ❑ Salvage value (disposal value)
- ❑ Cost basis (depreciation basis)
- ❑ Method of depreciation (how?)

Salvage value is the price of an equipment that can be obtained after it has been used.

What Can Be Depreciated?

- ✓ Assets used in business or held for production of income
 - ✓ Assets having a definite useful life and a life longer than one year
 - ✓ Assets that must wear out, become obsolete or lose value
-

A qualifying asset for depreciation must satisfy **all of the three conditions** above.

Cost Basis

Cost of a new hole-punching machine (Invoice price)	\$62,500
+ Freight	725
+ Installation labor	2,150
+ Site preparation	3,500
Cost basis to use in depreciation calculation	\$68,875

Depreciation Methods

Straight-Line Method
Declining Balance Method
Unit Production Method

Straight Line (SL) Method

This method assumes a uniform decrease in the value of asset with the passage of time.

- **Formula**

- **Annual Depreciation**

$$D_n = (I - S) / N, \text{ and constant for all } n.$$

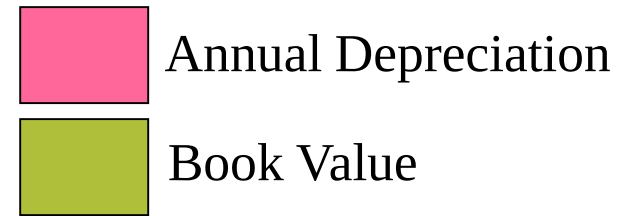
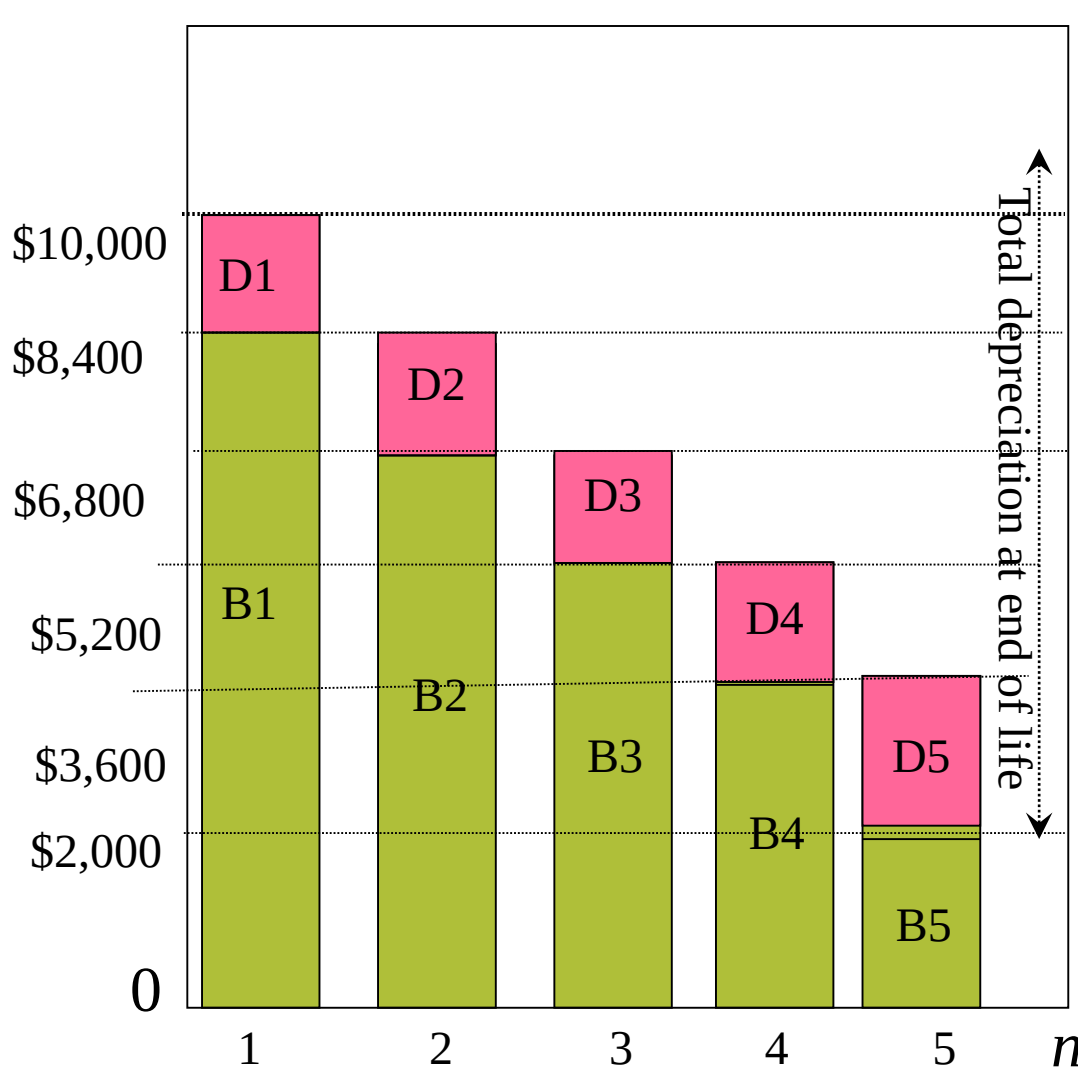
- **Book Value**

$$B_n = I - n (D)$$

where I = cost basis/value
 S = Salvage value
 N = depreciable life

Book value is the worth of an asset as shown on the accounting record of a company.

Example – Straight Line Method



$$I = \$10,000$$

$$N = 5 \text{ Years}$$

$$S = \$2,000$$

$$D = (I - S)/N$$

<i>n</i>	<i>D_n</i>	<i>B_n</i>
1	1,600	8,400
2	1,600	6,800
3	1,600	5,200
4	1,600	3,600
5	1,600	2,000

Declining Balance Method

In this method the depreciation cost is highest in the first year and reduces year after year.

Formula

- Annual Depreciation

$$D_n = \alpha B_{n-1} = \alpha I (1 - \alpha)^{n-1}$$

- Book Value

$$B_n = I(1 - \alpha)^n \quad \text{where } 0 < \alpha \leq 2(1/N)$$

Note: if α is chosen to be the upper bound, $\alpha = 2(1/N)$, we call it a 200% DB or double declining balance method.

Double declining method

This is one of the two common methods a company uses to account for the expenses of a fixed asset. This is an accelerated depreciation method. As the name suggests, it counts expense twice as much as the book value of the asset every year

Depreciation = $2 * \text{Straight line depreciation percent} * \text{book value at the beginning of the accounting period}$

Book value = Cost of the asset – accumulated depreciation

Example: On April 1, 2012, company X purchased an equipment for Rs. 100,000. This is expected to have 5 useful life years. The salvage value is Rs. 14,000. Company X considers depreciation expense for the nearest whole month. Calculate the depreciation expenses for 2012, 2013, 2014 using a declining balance method.

Useful life = 5

Straight line depreciation percent = $1/5 = 0.2$ or 20% per year

Depreciation rate = $20\% * 2 = 40\%$ per year

Depreciation for the year 2012 = $\text{Rs. } 100,000 * 40\% * 9/12 = \text{Rs. } 30,000$

Depreciation for the year 2013 = $(\text{Rs. } 100,000 - \text{Rs. } 30,000) * 40\% * 12/12 = \text{Rs. } 28,000$

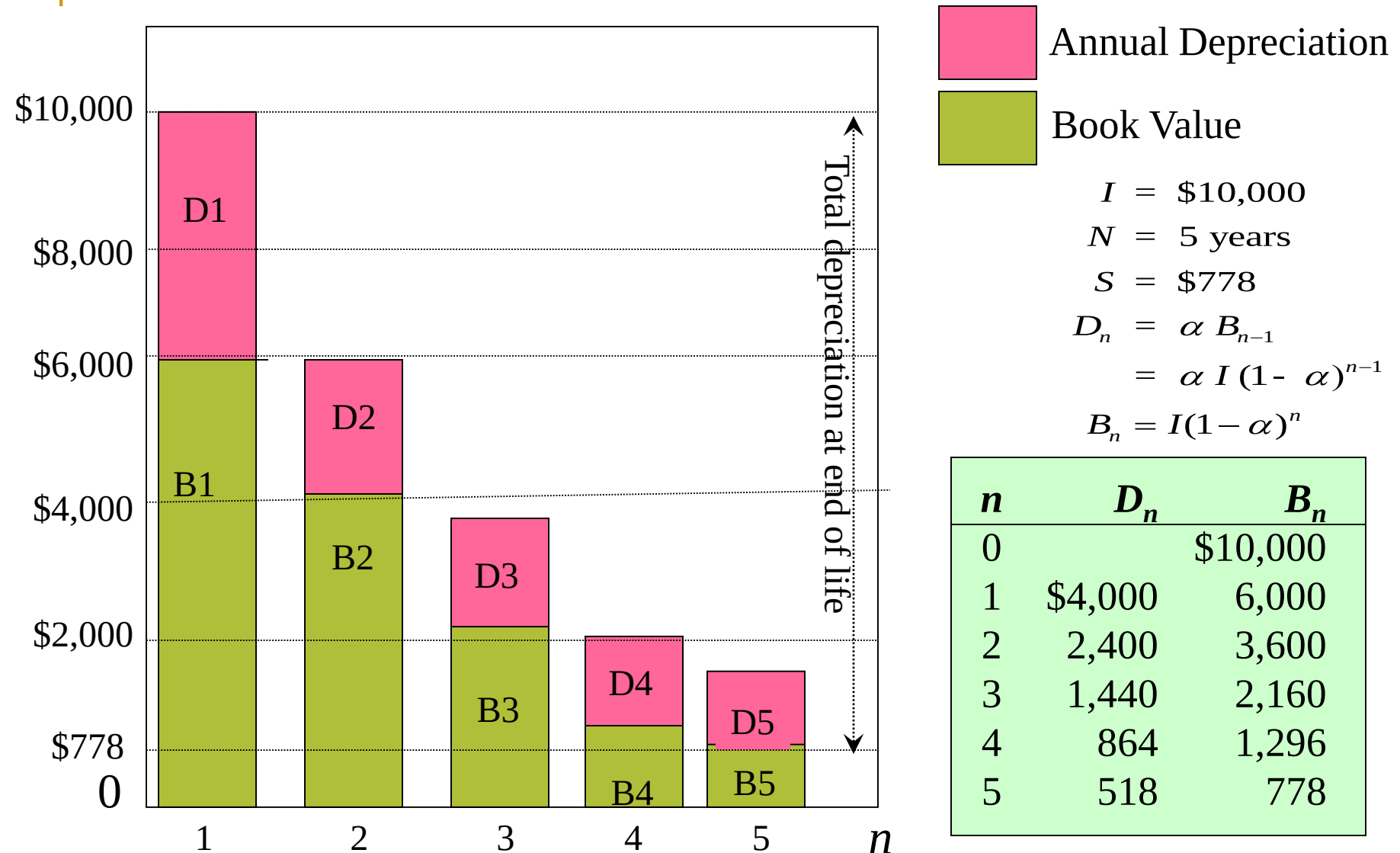
Depreciation for the year 2014 = $(\text{Rs. } 100,000 - \text{Rs. } 30,000 - \text{Rs. } 28,000) * 40\% * 12/12 = \text{Rs. } 16,800$

DDB

Depreciation table is shown below:

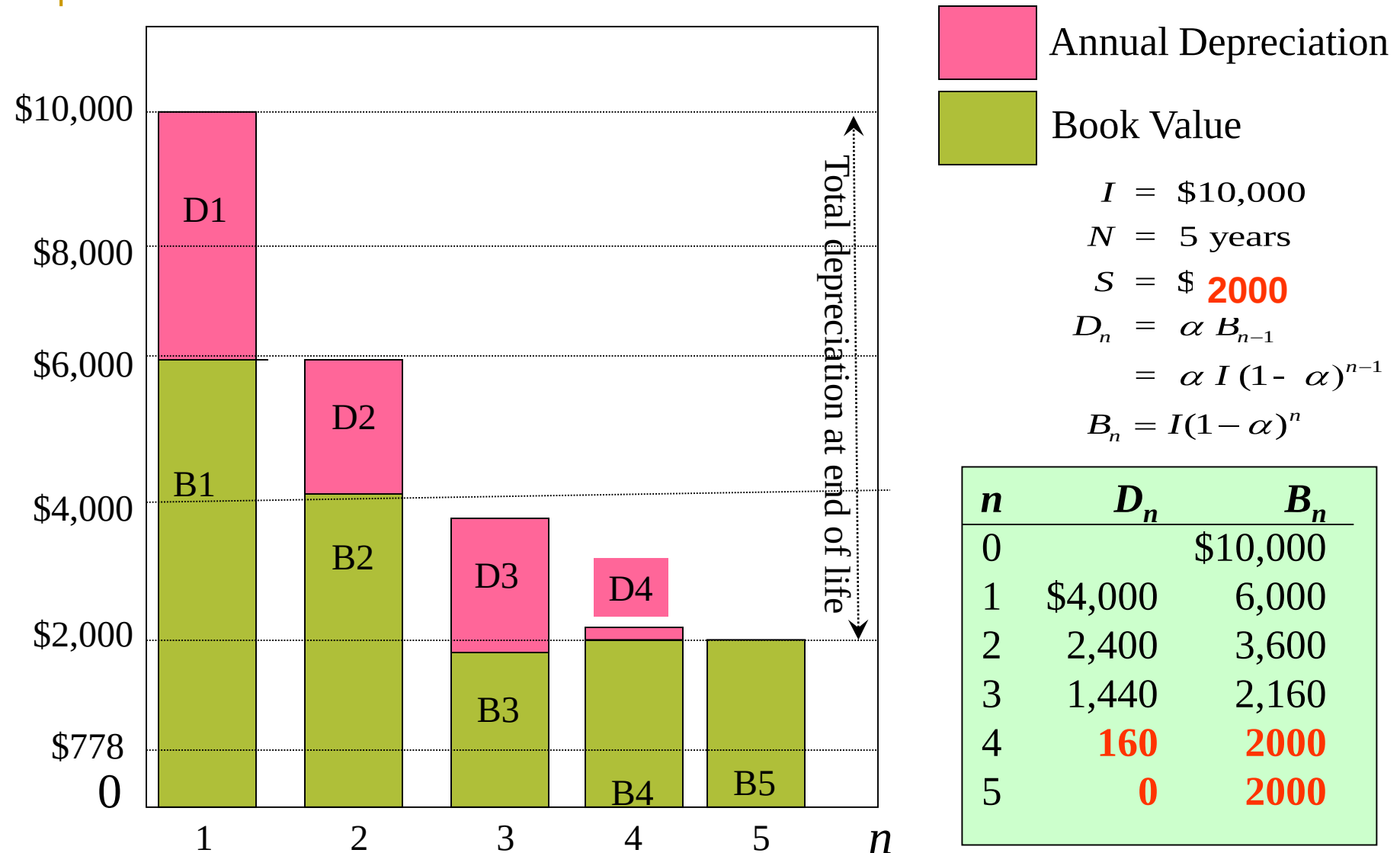
Year	Book value at the beginning	Depreciation rate	Depreciation Expense	Book value at the end of the year
2012	Rs. 100,000	40%	Rs. 30,000 * (1)	Rs. 70,000
2013	Rs. 70,000	40%	Rs. 28,000 * (2)	Rs. 42,000
2014	Rs. 42,000	40%	Rs. 16,800 * (3)	Rs. 25,200
2015	Rs. 25,200	40%	Rs. 10,080 * (4)	Rs. 15,120
2016	Rs. 15,120	40%	Rs. 1,120 * (5)	Rs. 14,000

Example – Declining Balance Method



$\alpha = .40$

Example – Declining Balance Method (if $B < \text{salvage value}$)



When $S = \$2,000$

End of Year	Depreciation	Book Value
1	$0.4(\$10,000) = \$4,000$	$\$10,000 - \$4,000 = \$6,000$
2	$0.4(6,000) = 2,400$	$6,000 - 2,400 = 3,600$
3	$0.4(3,600) = 1,440$	$3,600 - 1,440 = 2,160$
4	$0.4(2,160) = 864 > 160$	$2,160 - 160 = 2,000$
5	0	$2,000 - 0 = 2,000$

Note: Tax law does not permit us to depreciate assets below their salvage values.

Units-of-Production Method

- **Principle**

Service units will be consumed in a non time-phased fashion

- **Formula**

- Annual Depreciation

$$D_n = \frac{\text{Service units consumed for year}}{\text{total service units}} (I - S)$$

Example

- **Given:** $I = \$55,000$, $S = \$5,000$, Total service units = 250,000 miles, usage for this year = 30,000 miles
- **Solution:**

$$\begin{aligned} Dep &= \frac{30,000}{250,000} (\$55,000 - \$5,000) \\ &= \left(\frac{3}{25} \right) (\$50,000) \\ &= \$6,000 \end{aligned}$$

Discussion